

Problem-Solution Fit

Date	27 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience .

1. CUSTOMER SEGMENT(S) [CS] Disaster management authorities, local government agencies, emergency response teams, and people living in flood-prone areas.	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] Limited access to accurate flood prediction systems, lack of technical expertise, and dependence on publicly available weather information.	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Pros: Weather forecasting services provide rainfall updates and alerts. Cons: They do not always provide accurate, localized flood risk predictions or probability estimates.
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] • Delayed flood prediction increases the risk of loss of life and property. • Floods occur frequently during heavy rainfall and monsoon seasons.	9. PROBLEM ROOT / CAUSE [RC] Heavy rainfall, poor drainage systems, rising river discharge, and climate change contribute to increased flood risk.	7. BEHAVIOR + ITS INTENSITY [BE] Users regularly monitor weather forecasts during monsoon seasons and seek quick, reliable flood prediction before taking precautionary measures.
3. TRIGGERS TO ACT [TR] Heavy rainfall, increasing river discharge, abnormal weather conditions, and flood warnings trigger users to check flood risk.	10. YOUR SOLUTION [SL] A Flask-based Machine Learning application that analyzes weather parameters and predicts flood probability and risk level (Low, Moderate, High, Severe) to support early warning and disaster preparedness.	8. CHANNELS OF BEHAVIOR [CH] ONLINE: Weather websites, mobile apps, government portals. OFFLINE: Television, radio, newspapers, local authorities, disaster management agencies.
4. EMOTIONS BEFORE / AFTER [EM] Before: Fear, uncertainty, and lack of preparedness. After: Confidence, awareness, and better preparedness for possible flooding.		